

The Distributive Property

Lesson 6-5

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$$3(x + 2)$$

$$3x + 6$$

$3(4 + 5) = 3 \times 4 + 3 \times 5 = 12 + 15 = 27$ — the 3 gets 'distributed' to both the 4 and the 5

Distributive Property

Multiplying a number by everything inside the parentheses: $a(b + c) = ab + ac$.

$$3 \times 4 = 12$$

3 and 4 are factors

In $3(x + 2)$, the 3 is the factor outside the parentheses that multiplies each term inside

Factor

A number that gets multiplied by another number.

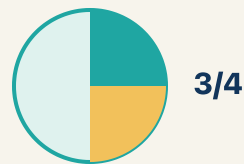
$$3(x + 2)$$

$$3x + 6$$

Expand $2(n + 6)$: multiply $2 \times n = 2n$ and $2 \times 6 = 12$, so $2(n + 6) = 2n + 12$

Expand

To multiply out the parentheses in an expression.



$3(x + 4)$ and $3x + 12$ always give the same answer: when $x = 2$, both = 18; when $x = 10$, both = 42

Equivalent

Expressions that always have the same value.


$$4x$$

coefficient = 4

In $6x + 15$, the coefficient of x is 6 — it came from distributing in $3(2x + 5)$

Coefficient

The number in front of a letter, like the 3 in $3x$.


$$3x + 2x = 5x$$

same variable → combine

$4x + 2x = 6x$ (combine coefficients); $4x + 2y$ cannot be combined (different variables)

Like terms

Terms with the same letter, like $2x$ and $5x$.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can use the distributive property to expand and factor expressions.

1 Notice

You're selling concert bundles at the music studio.

2 Key idea

I can use the distributive property to expand and factor expressions.

3 Words I'll use

Distributive Property

Factor

Expand

Equivalent

Coefficient

Like terms



Think About It

- Why do both methods give the same total?
- What does the 3 represent in $3(15 + 5)$?
- How is $3(15 + 5)$ related to $3 \times 15 + 3 \times 5$?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: Expand $6(n + 3)$ using the distributive property.

A. $6n + 18$

B. $6n + 3$

C. $n + 18$

D. $6n + 9$

Step 1 $6(n + 3) = 6 \times n + 6 \times 3 = 6n + 18.$



Answer: A. $6n + 18$

 We do — together

Solve this one with your class using the same steps.

Problem: Expand $4(5 - 2)$ using the distributive property.

- A. 12
- B. $20 - 2$
- C. 9
- D. 22

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: Expand $3(x + 4)$ using the distributive property.

- A. $3x + 12$
- B. $3x + 4$
- C. $x + 12$
- D. $7x$

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Each bundle is a ticket (\$15) and a snack (\$5). For 3 people, why does $3(15 + 5)$ give the same \$60 as $3 \times 15 + 3 \times 5$?

Level 1 support

Start here: The distributive property multiplies the 3 by each item, giving the same total.

 **Need a hint?**

Sentence starters (optional)

Both give \$60 because I distribute the 3 to ____ and ____.

Las dos dan \$60 porque distribuyo el 3 a ____ y ____.

So $3(15 + 5)$ equals ____ because ____.

Entonces $3(15 + 5)$ es igual a ____ porque ____.

Word bank: Distributive Property Factor Expand Equivalent

Listen for: Students explain the 3 distributes to both the ticket and the snack ($3 \times 15 + 3 \times 5 = 45 + 15 = 60$), matching 3×20 , and use the word equivalent.

Level 2

If 10 people bought bundles, which form is faster to compute: $10(15 + 5)$ or $10 \times 15 + 10 \times 5$? Justify your choice.

- The ____ form is faster because ____.
- Either way the total is ____, since both forms are equivalent because ____.

EXPLORE

When you expand $5(2x + 4)$, why must the 5 multiply BOTH terms inside the parentheses?

Level 1 support

Start here: The factor outside multiplies every term inside the parentheses.

 **Need a hint?**

Sentence starters (optional)

I distribute the 5 to ____ and ____, so $5(2x + 4) =$ ____.

Distribuyo el 5 a ____ y ____, así $5(2x + 4) =$ ____.

Both terms get multiplied because ____.

Los dos términos se multiplican porque ____.

Word bank: **Distributive Property** **Expand** **Factor** **Equivalent**

Listen for: A strong answer expands $5(2x + 4) = 10x + 20$ and explains the parentheses mean 5 copies of the whole sum, so both $2x$ and 4 are multiplied.

Level 2

A classmate writes $5(2x + 4) = 10x + 4$. Critique the error and prove the correct form is equivalent by testing $x = 3$.

- The error is ____ because they only multiplied ____.
- Testing $x = 3$: $5(2 \cdot 3 + 4) =$ ____ but $10 \cdot 3 + 4 =$ ____, so they are not equivalent.

CONNECT

For 8 practice rooms each needing a music stand (\$12) and a stool (\$9), the cost is $8(12 + 9)$. How do you use the distributive property to find the total, and which form is easier for you?

Level 1 support

Start here: Distribute the 8 to both items, then add.

 **Need a hint?**

Sentence starters (optional)

$$8(12 + 9) = 8 \times \underline{\quad} + 8 \times \underline{\quad} = \underline{\quad}.$$

$$8(12 + 9) = 8 \times \underline{\quad} + 8 \times \underline{\quad} = \underline{\quad}.$$

I think the _____ form is easier because _____.

Creo que la forma _____ es más fácil porque _____.

Word bank: Distributive Property Factor Expand Equivalent

Listen for: Students compute $8 \times 12 + 8 \times 9 = 96 + 72 = 168$ (or $8 \times 21 = 168$) and justify which form they find easier, recognizing both are equivalent.

Level 2

Working backwards, how would you factor $8x + 12$ into a product? Explain how factoring undoes the distributive property.

- I factor out the _____, giving _____.
- Factoring undoes expanding because _____.

Write About the Math

The Writing Revolution

I can explain my steps using the words distributive property, factor, expand, and equivalent.

1. Kernel Sentence subject + verb

Model: Distributive Property is multiplying a number by everything inside the parentheses: $a(b + c) = ab + ac$.

Propiedad distributiva es multiplicar un número por todo lo que está dentro del paréntesis: $a(b + c) = ab + ac$.

Write a kernel sentence about distributive property. Use a subject and a verb.

Escribe una oración base sobre propiedad distributiva. Usa un sujeto y un verbo.

2. Sentence Expansion because • but • so

Kernel: Distributive Property matters in math

Propiedad distributiva importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Distributive Property matters in math because ____.

Propiedad distributiva importa en matemáticas porque ____.

but
pero

Distributive Property matters in math, but ____.

Propiedad distributiva importa en matemáticas, pero ____.

so
entonces

Distributive Property matters in math, so ____.

Propiedad distributiva importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about distributive property.
Di un hecho verdadero sobre distributive property.

Distributive property ____.

Question *Pregunta*

Ask a question about distributive property.
Haz una pregunta sobre distributive property.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about distributive property.
Muestra entusiasmo sobre distributive property.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with distributive property.
Dile a un compañero qué hacer con distributive property.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

Both give \$60 because I distribute the 3 to ____ and ____.

Las dos dan \$60 porque distribuyo el 3 a ____ y ____.

So $3(15 + 5)$ equals ____ because ____.

Entonces $3(15 + 5)$ es igual a ____ porque ____.

I distribute the 5 to ____ and ____, so $5(2x + 4) =$ ____.

Distribuyo el 5 a ____ y ____, así $5(2x + 4) =$ ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Which expression is equal to $6x + 18$ using a common factor?

- A. $6(x + 3)$
- B. $6(x + 18)$
- C. $x(6 + 18)$
- D. $6(x + 12)$

Show your work:

2. Use the Distributive Property to compute 7×104 .

- A. 728
- B. 700
- C. 98
- D. 742

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Use the distributive property to explain why 6×98 can be calculated as $6 \times 100 - 6 \times 2 = 588$. Then create your own example of using the distributive property to make multiplication easier.

Sentence starter: $6 \times 98 = 6(100 - 2) = 6 \times \underline{\quad} - 6 \times \underline{\quad} = \underline{\quad} - \underline{\quad} = \underline{\quad}$. *My example:* $\underline{\quad} \times \underline{\quad} = \underline{\quad}(\underline{\quad} \pm \underline{\quad}) = \underline{\quad}$.

Show your work:

Reflect — Exit Ticket

Which expression is equivalent to $7(x + 3)$?

- A. $7x + 21$
- B. $7x + 3$
- C. $x + 21$
- D. $7x + 10$

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. $12 - 4(5 - 2) = 4 \times 5 - 4 \times 2 = 20 - 8 = 12$.
2. **You Try (notes):** A. $3x + 12 - 3(x + 4) = 3 \times x + 3 \times 4 = 3x + 12$.
3. **Try It 1:** A. $6(x + 3)$ — *The GCF of 6x and 18 is 6: $6(x + 3)$.*
4. **Try It 2:** A. $728 - 7 \times 104 = 7 \times 100 + 7 \times 4 = 700 + 28 = 728$.
5. **Exit Ticket:** A. $7x + 21 - 7(x + 3) = 7 \times x + 7 \times 3 = 7x + 21$.

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Distributive Property is multiplying a number by everything inside the parentheses: $a(b + c) = ab + ac$.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).